

A Reflective Journey: Navigating Your Cumulative Experience at Iowa State University

When I first started at Iowa State as a Cybersecurity Engineer, I really didn't know what it meant to be an engineer. I assumed that this program would give me a deeper technical understanding of computers, networks, software, and Cybersecurity. At the time, I did not know how engineering fit into this degree. After spending over three years in the program, I have a far better understanding of what an engineer is. Engineering is about solving problems, not just knowing technical information. I now see that being an engineer means taking a problem that may not have an obvious answer and working through it step by step. Often, the hardest part is not solving the problem, but figuring out what the real problem is. To make things even more complex, you need to take into consideration cost, time, security, privacy, usability, ethics, societal impact, your users' needs, and more. A solution can end up being technically excellent, but still be an awful engineering solution because of extensive cost, negative societal impact, unethical outcomes, etc. Another important piece of the puzzle is teamwork. Having the right answer is not useful if you cannot communicate that answer to other engineers, business stakeholders, students, or users. Engineering is a collaborative field, meaning that the best solutions stem from groups of people, not one individual. The biggest lesson I have learned is that being an engineer does not mean having every answer. It means having the foundation and willingness to research, ask questions, adapt, and learn when you do not.

My coursework at Iowa State has taught me to think like an engineer. Cybersecurity Engineering has forced me to understand multiple connected areas of technology, instead of just one isolated area. I have learned about programming, networking, operating systems, computer systems, and Cybersecurity. As I progressed through the program, I started to understand how connected all of these concepts really are. A decision made in one part of a system can create problems somewhere else. For example, in networking, knowing just one internet protocol by itself is not enough. You need an understanding of how data moves through

the entire network and how each part of that process affects the next. Another good example is operating systems, where I learned that synchronization, memory management, and protection all depend on different parts of the system working together. These classes have helped me become more confident when approaching technical problems. I am now able to break a complex problem into smaller pieces, understand how those pieces interact, and then look at the system as a whole. This has caused a major shift in how I approach problems. I am no longer simply searching for the “correct” answer, but rather focusing on finding a solution that actually works within the complex system.

Another experience that has greatly changed my understanding of engineering has been serving as a Teaching Assistant for the Security Ethics course at Iowa State. Teaching this course has forced me to look beyond the technical side of Cybersecurity and consider the ethical responsibilities that come with being an engineer. Cybersecurity professionals constantly make decisions involving privacy, security, transparency, individual rights, and organizational needs. There is rarely one clear-cut answer, and just because something can be done technically does not always mean that it should be. Teaching these concepts has also improved my communication skills. Each student thinks about a problem differently, so I have learned to adapt the way I communicate depending on my audience. This experience has also shown me that teaching is one of the best ways to test my own understanding. If I cannot explain something clearly to another person, there is a good chance I do not understand it as well as I think I do.

Some of my most valuable learning has happened outside of the classroom through my internship at Kingland Systems as an Advanced Cybersecurity Governance, Risk, and Compliance Analyst. At Kingland, I worked alongside the full-time Cybersecurity team conducting risk assessments, participating in SOC 2 and ISO audits, developing policies, and rebuilding the company’s ethics reporting hotline. Working with ISO standards exposed me to

the global side of cybersecurity because these standards are used internationally. The biggest difference between this experience and the classroom was that I was not always given a clearly defined problem or all of the information needed to solve it. I had to use standards, internal documentation, professional resources, and the experience of my coworkers to find the information I needed. Working with audits taught me that understanding a security control goes far beyond simply reading the requirement. I had to understand why the control existed, what risk it addressed, and what evidence showed that it was actually working. Rebuilding Kingland's internal ethics reporting hotline was another example where there was no step-by-step solution. I had to consider confidentiality, employee trust, access, governance, and how the system would actually be used. After plenty of trial and error, my work was rolled out company-wide, and gave our HR department an easier way to track, review, and respond to internal reports. These experiences taught me that real engineering problems require you to use what you already know, identify what you are missing, and adapt as you learn more.

My approach to learning has changed significantly throughout my time at Iowa State as well. Earlier in college, I relied heavily on lecture slides, notes, and practicing problems that were similar to ones I had already seen, leading me to memorize how to solve one type of problem. As my courses became more difficult, I realized that memorizing how to solve one type of problem was not going to be enough. I started focusing more on why something worked instead of simply memorizing the steps to get the right answer. I now break problems into smaller pieces, use outside resources when something does not make sense, ask more questions, and try to connect new concepts to things I already understand. If I could restart my undergraduate experience knowing what I know now, I would probably focus less on learning material just for the next exam and spend more time trying to understand how it could be applied outside of the classroom. Grades are still important to me, but I have learned that getting a good grade and truly understanding something are not always the same thing. I would

also be quicker to ask questions and use the knowledge of professors, classmates, and professionals instead of feeling like I need to figure everything out on my own.

Looking forward, I know that my learning will not stop when I graduate from Iowa State. Cybersecurity changes far too quickly for the information I know today to carry me through an entire career. Technologies will change, new threats will emerge, regulations will evolve, and I will constantly be faced with problems that I have never seen before. As I continue into my career, I want to deepen both my technical Cybersecurity knowledge and my understanding of Governance, Risk, and Compliance. I also want to continue improving as a communicator and leader, especially when it comes to explaining technical risks to people without a technical background. When I first came to Iowa State, I thought the goal was to graduate knowing enough information to become an engineer. After more than three years, I have realized that this is not really possible. Iowa State cannot teach me everything I will ever need to know, nor should it. Instead, it has given me the technical foundation, problem-solving skills, and confidence to continue learning when I am faced with something I do not already know. To me, that is one of the most important things it means to be an engineer.